Two Positive-Increase Endpoint Questions from arXiv:1305.5365 Are Answered by arXiv:1709.08895

FA/Banach run literature-status packet

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Status

This is a `literature_already_answered` packet for two scoped questions in Batty–Chill–Tomilov [1]. Rozendaal–Seifert–Stahn [2] explicitly extend the relevant source results to every resolvent control function of positive increase. This is a literature identification, not a new proof from the run.

The two source questions

On PDF page 5, immediately after Theorem 1.3(b), [1] considers a bounded C_0-semigroup $(T(t))_{t\geq 0}$ on a Hilbert space, generated by $-A$, with $\sigma(A)\cap i\mathbb{R}=\varnothing$. For $\alpha>0$ and $\beta\geq 0$, the paper proves that

$$\|(is+A)^{-1}\| = O\Big(|s|^{\alpha}(\log|s|)^{\beta}\Big)$$

implies, for every $\varepsilon>0$,

$$\|T(t)A^{-1}\| = O\Big(t^{-1/\alpha}(\log t)^{\varepsilon+\beta/\alpha}\Big).$$

It then asks whether this is true with $\varepsilon = 0$.

On PDF page 6, immediately before Theorem 1.5, the source says that its zero-frequency theorem covers resolvent growth slightly slower than a power of $|s|^{-1}$, while the corresponding characterization for growth slightly faster than a power remains open.

The later answer

The later paper [2] cites [1] as reference [6]. On PDF page 3 it says that the source obtains the improved estimate only for a subclass of regularly varying functions and announces the extension to *all* functions M having positive increase.

Theorem 3.2 of [2] (PDF page 8) states that, if $\|R(is,A)\|\leq M(|s|)$ and M is continuous, non-decreasing, and of positive increase, then

$$\|T(t)A^{-1}\| = O\left(\frac{1}{M^{-1}(t)}\right).$$

Every regularly varying function of positive index has positive increase ([2], PDF page 6). Hence this applies to

$$M(s) = s^{\alpha}(\log s)^{\beta}.$$

Since

$$M^{-1}(t) \asymp t^{1/\alpha}(\log t)^{-\beta/\alpha},$$

the conclusion is precisely

$$\|T(t)A^{-1}\| = O\Big(t^{-1/\alpha}(\log t)^{\beta/\alpha}\Big).$$

Thus the requested $\varepsilon = 0$ endpoint is affirmative.

Theorem 3.6 of [2] (PDF page 13) gives the zero-frequency analogue. If $\sigma(A) \cap i\mathbb{R} = \{0\}$, the resolvent is bounded for $|s| \geq 1$, and

$$\|R(i/s, A)\| \leq M(|s|), \qquad |s| \geq 1,$$

for a positive-increase function M, then

$$\|T(t)AR(1, A)\| = O\left(\frac{1}{M^{-1}(t)}\right).$$

Taking $M(s) = s^{\alpha}(\log s)^{\beta}$ supplies exactly the missing faster-than-a-power upper bound. With the minimal resolvent envelope, or matching two-sided growth, the source's lower bounds give the corresponding sharp characterization.

Scope limitation

This note does not settle Remark 4.8 of [1], concerning an arbitrary bounded operator B commuting with the semigroup. The implication

$$\|T(t)B\| = O(t^{-1}) \quad \Longrightarrow \quad \sup_{\operatorname{Re}\lambda > 0} \|B(\lambda + A)^{-1}\| < \infty$$

is a separate strict endpoint. Eight direct and literature routes were audited in the run's companion attempt record without obtaining a proof or counterexample.

References

[1] C. J. K. Batty, R. Chill, and Y. Tomilov, *Fine scales of decay of operator semigroups*, arXiv:1305.5365; J. Eur. Math. Soc. 18 (2016), 853–929.

[2] J. Rozendaal, D. Seifert, and R. Stahn, *Optimal rates of decay for operator semigroups on Hilbert spaces*, arXiv:1709.08895; Adv. Math. 346 (2019), 359–388.